

Q6		
(a)(i)	The resistance is infinite.	B1
(a)(ii)	The resistance decreases as V increases.	B1
(b)(i)	Method 1: When the galvanometer reads zero, $V_{XZ} = V_{LDR} = \frac{R_{LDR}}{R_{LDR} + R} E = \frac{1600}{1600 + 1200} (9.0) = 5.143 \text{ V}$ For the wire, $\frac{V_{XZ}}{V_{ZY}} = \frac{L_{XZ}}{L_{ZY}} \Rightarrow L_{XZ} = \frac{V_{XZ}}{V_{ZY}} L_{ZY} = \frac{5.143}{9.0} (1.2) = 0.6857$ $= 0.69 \text{ m}$ Method 2: $\frac{V_{XZ}}{V_{ZY}} = \frac{V_R}{V_{LDR}} \Rightarrow \frac{kL_{XZ}}{kL_{ZY}} = \frac{I \times 1600}{I \times 1200} = \frac{4}{3}$ $L_{XZ} = \frac{4}{7} \times 1.2$ $= 0.69 \text{ m}$	M1 A1 M1 A1
(b)(ii)1	As intensity of light is increased, the resistance of the LDR decreases and there is a smaller potential difference across the LDR. The length of XZ decreases.	M1 A1
(b)(ii)2	The total resistance of the circuit decreases and more current is drawn from the battery. Hence power produced by the battery increased. OR The total resistance of the circuit decreases. Since power produced by battery = V^2/R_{total}, power produced by battery increased.	M1 A1 M1 A1

07		
----	--	--